

System Specification: End-to-End Flow Diagram & Data Lifecycle

1. System Architecture & Workflow Diagram

This document illustrates the complete end-to-end operational flow of the EQUAL system—from initial multi-format file upload through AI narrative extraction, job monitoring, quarantine handling, master schema standardization, change history management, matrix heatmap visualization, and maintenance plan (MP) versioning.



STEP 8

ISO Date & Equipment Alias Standardization

**STEP 9**

Row Verification Check

*Valid Report Content / Blank / Noise / Non-Content***STEP 10**

Jobs Execution Queue

**STEP 11**

Quarantine Subsystem

**STEP 12**

Jobs Table 4s Scoped Polling

*Eye Icon Click***STEP 13**

Job Preview Modal 22 Master Attributes Preview

STEP 14

Quarantine Table WHY Reason Pills

*Restore Action***STEP 15**

Re-inject Row into Active Pool

**STEP 16**

Change History Data Grid

**STEP 17**

Header Actions: Import CSV / Export CSV & Excel

STEP 18

Targeted ZIP Export: POST /api/ChangeData/ExportZipByIds

STEP 19

Save Changes: POST /api/ChangeData/SaveChangedData



Auto-Trigger

STEP 20

Event Listener: refreshChangeHistoryData



STEP 21

Matrix Inquiry 2D Heatmap & Photo Gallery

STEP 22

MP List Inquiry Version Control & Row Revisions

STEP 23

MP List Management Spec Column Sequence Engine

2. Stage-by-Stage Operational Summary for QC Testing

Stage #	Stage Name	System Action	QC Verification Expectation
1	Multi-Format Ingestion	Ingests .xlsx , .csv , .pdf , .pptx , .png , .eml files.	System accepts all 5 supported file types without rejection.
2	Pre-Upload Report	Displays column headers and editable prompt with Change Points.	Column summary tags match input file headers exactly.
3	AI Extraction	GPT Opus 120B maps text to 22 master schema columns.	Jargon normalized; dates converted to ISO-8601 (YYYY-MM-DD).
4	Routing & Isolation	Valid rows sent to Jobs; blank/noise sent to Quarantine.	Invalid/empty rows isolated without breaking the job run.
5	Jobs & Quarantine	4s polling on Jobs; Restore action on Quarantine.	Polling active only on Jobs page; Restore returns row to pool.
6	Change History Data	22-attribute grid editing, targeted ZIP export by IDs.	Selected checkbox IDs exported in ZIP package.

Stage #	Stage Name	System Action	QC Verification Expectation
7	Matrix & MP Modules	Heatmap status badges, MP versioning, column sequence.	Column order strictly matches sequence numbers.